E-Commerce Sales & Customer Analysis

Complete Exploratory Data Analysis Report

Prepared by: Gayatri Deshmukh

Project Overview

Analyze sales, profitability, customers, regions, discounts, payment methods, and order statuses using a reproducible Python data analysis pipeline.

Dataset

Global E-Commerce Sales Dataset, 2021–2024. Synthetic data for learning and portfolio analysis.

Source: https://www.kaggle.com/datasets/abdelfattahibrahim/global-e-commerce-sales-dataset-20212024

Clean records: 10,000. Date range: 05 January 2021 to 24 December 2024. Products: 70. Categories: 5. Regions: 4.

Financial analysis scope: All order statuses.

Financial amounts are shown in source currency units. Customer names are unavailable; customer IDs are used.

Tools and Workflow

Python, Pandas, NumPy, Matplotlib, Seaborn, and ReportLab. Workflow: raw CSV → inspection → cleaning → transformation → analysis → charts → business findings → PDF report.

Data Cleaning Summary

Cleaning check	Count
Input rows	10,000
Exact duplicates removed	0
Missing or invalid essential rows removed	0
Invalid transaction rows removed	0
Categorical values filled	0
Missing discount filled with zero	0
Missing profit filled with zero	0
Invalid discount rows removed	0
Duplicates after standardization removed	0
Output rows	10,000
Total rows removed	0

Cleaning trims text, parses dates, converts numeric fields, removes exact duplicates and invalid essential transactions, and validates discount rates. Negative profits and valid outliers are retained.

Missing profit or discount cells are filled with zero only under the documented instructor assumption. Refer to the audit counts above to determine whether this affected results.

Business KPI Summary

Metric	Value
Recorded revenue	5,284,387.70
Recorded profit	1,437,638.31
Unique orders analyzed	10,000
Known customers analyzed	5,348
Average order value	528.44
Overall profit margin	27.21%
Cancelled orders, all records	908
Returned orders, all records	1,857

AOV = total recorded revenue divided by distinct analyzed order IDs. Overall profit margin = total recorded profit divided by total recorded revenue, multiplied by 100.

Status counts cover all clean records. Financial totals follow the stated analysis scope. When all statuses are included, these totals are not delivered-only or refund-adjusted results.

Key Findings

Highest-revenue region: Middle East (1,348,593.22).

Highest-revenue category: Electronics (3,382,028.46).

Highest total-profit category: Electronics (925,448.15).

Highest-revenue product: Lenovo ThinkPad X1 (379,258.55).

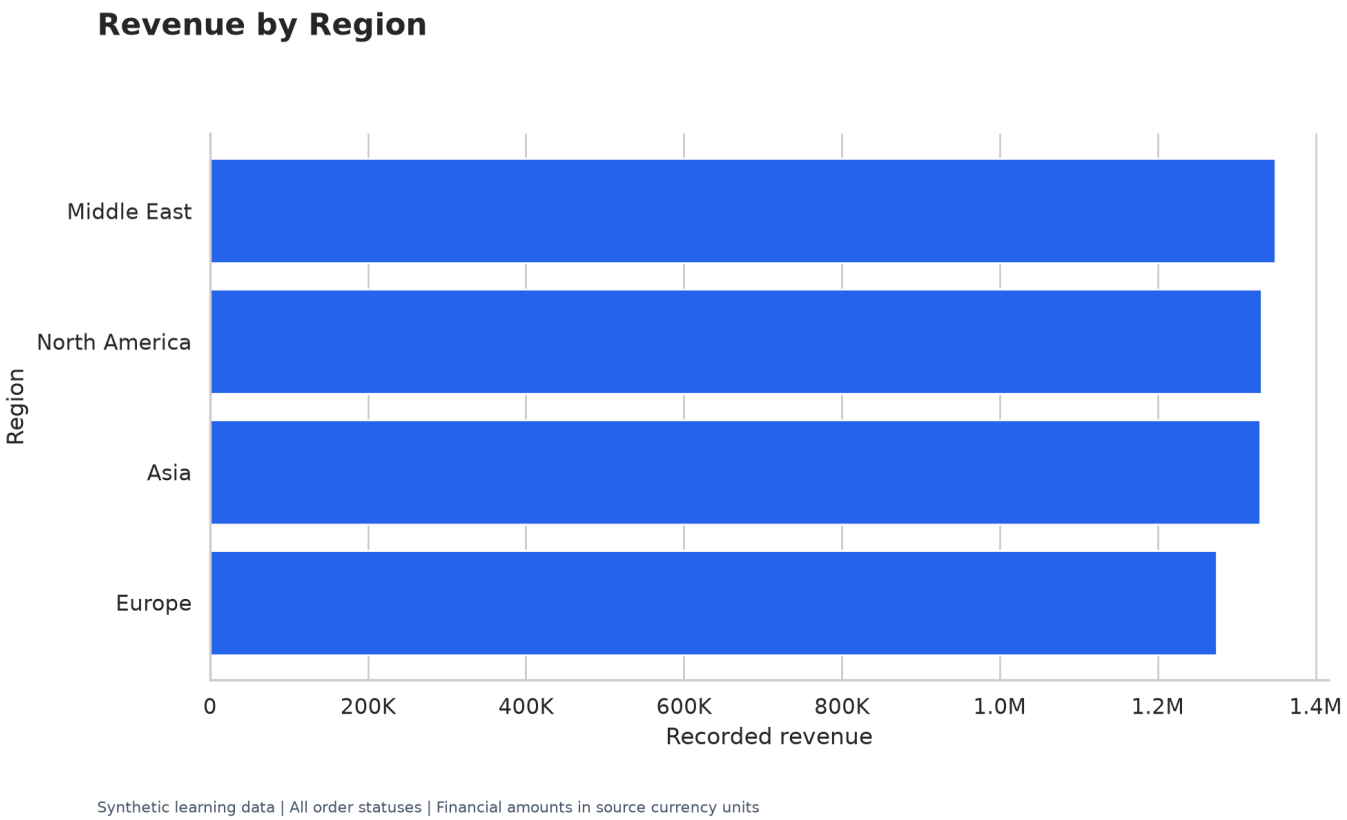
Highest-revenue customer: CUST-01046 (20,094.80).

Highest recorded monthly revenue: 2023-10 (243,996.98).

Most-used payment method: Debit Card (1,486 distinct orders).

Discount and row profit-margin correlation: -0.83. This association does not establish causation.

Revenue by Region

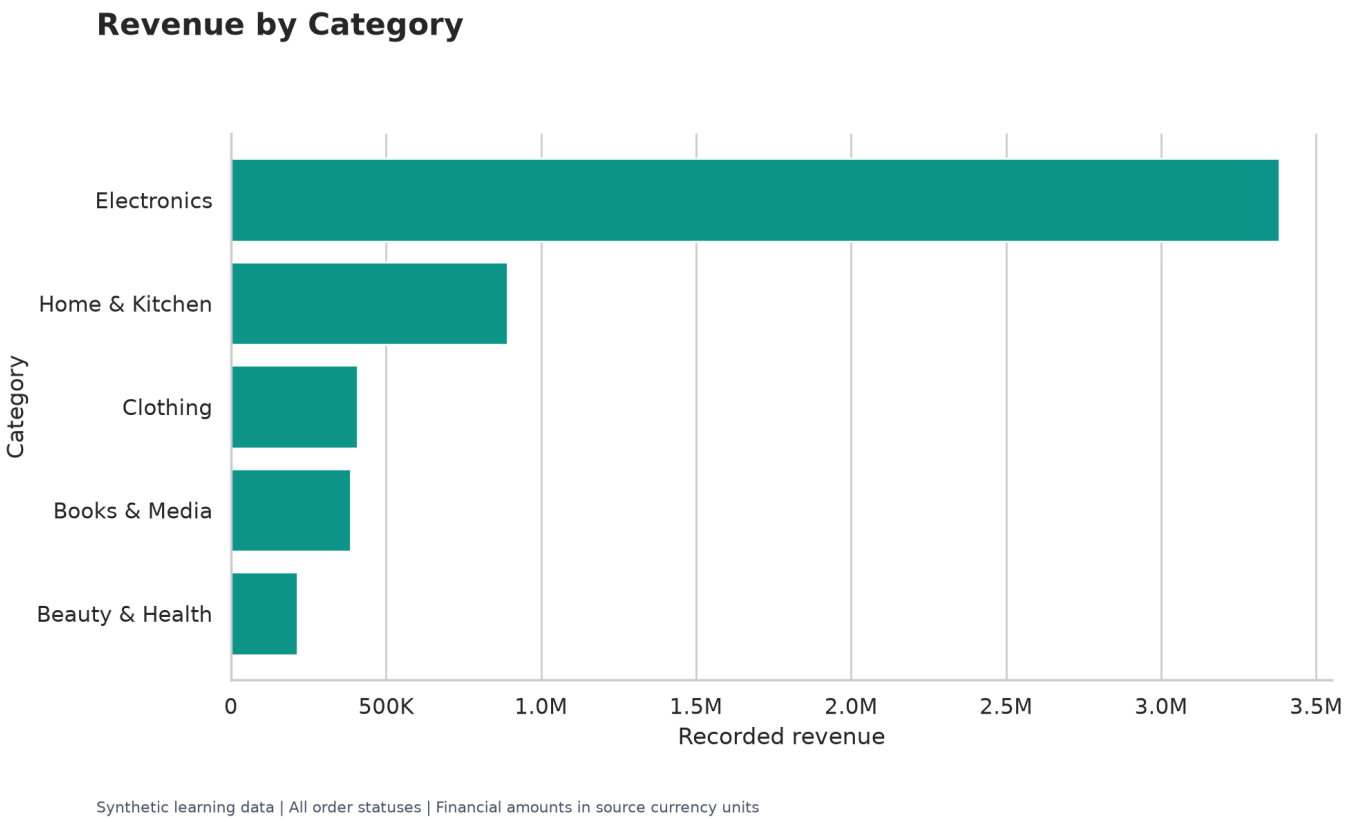


Synthetic learning data | All order statuses | Financial amounts in source currency units

Compare recorded revenue across regions. Revenue differences may reflect order counts, product mix, prices, or quantities; this chart alone does not explain their causes.

Analysis scope: All order statuses.

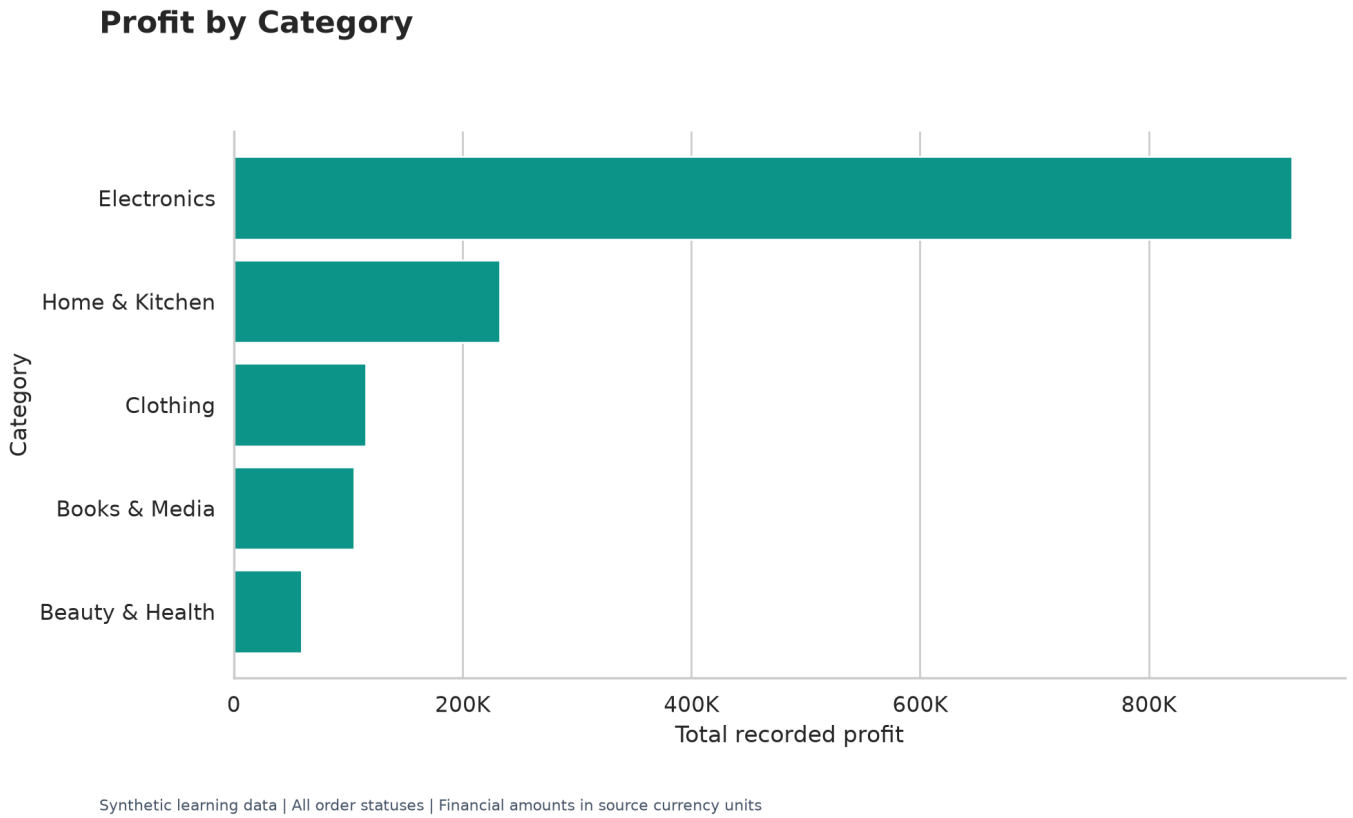
Revenue by Category



Compare category contributions to recorded revenue. High revenue does not necessarily imply a high profit margin.

Analysis scope: All order statuses.

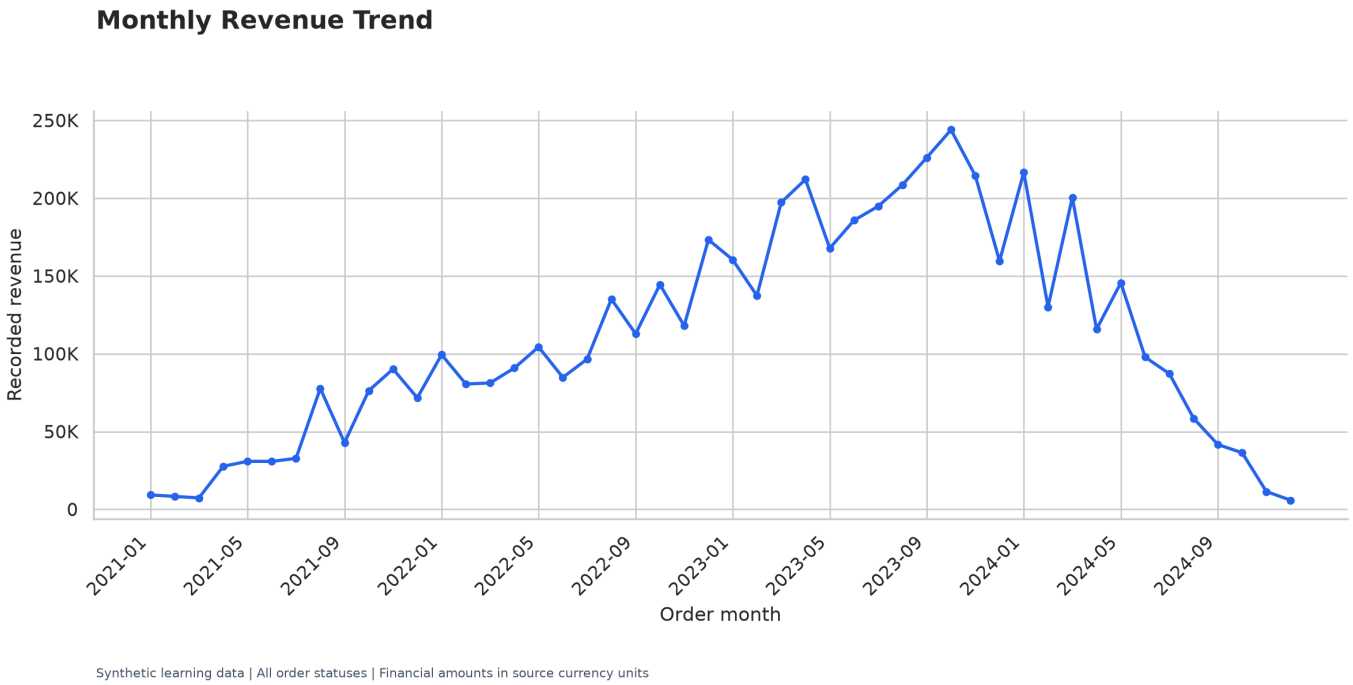
Profit by Category



Compare total profit contributions. Assess these alongside category revenue and profit margins.

Analysis scope: All order statuses.

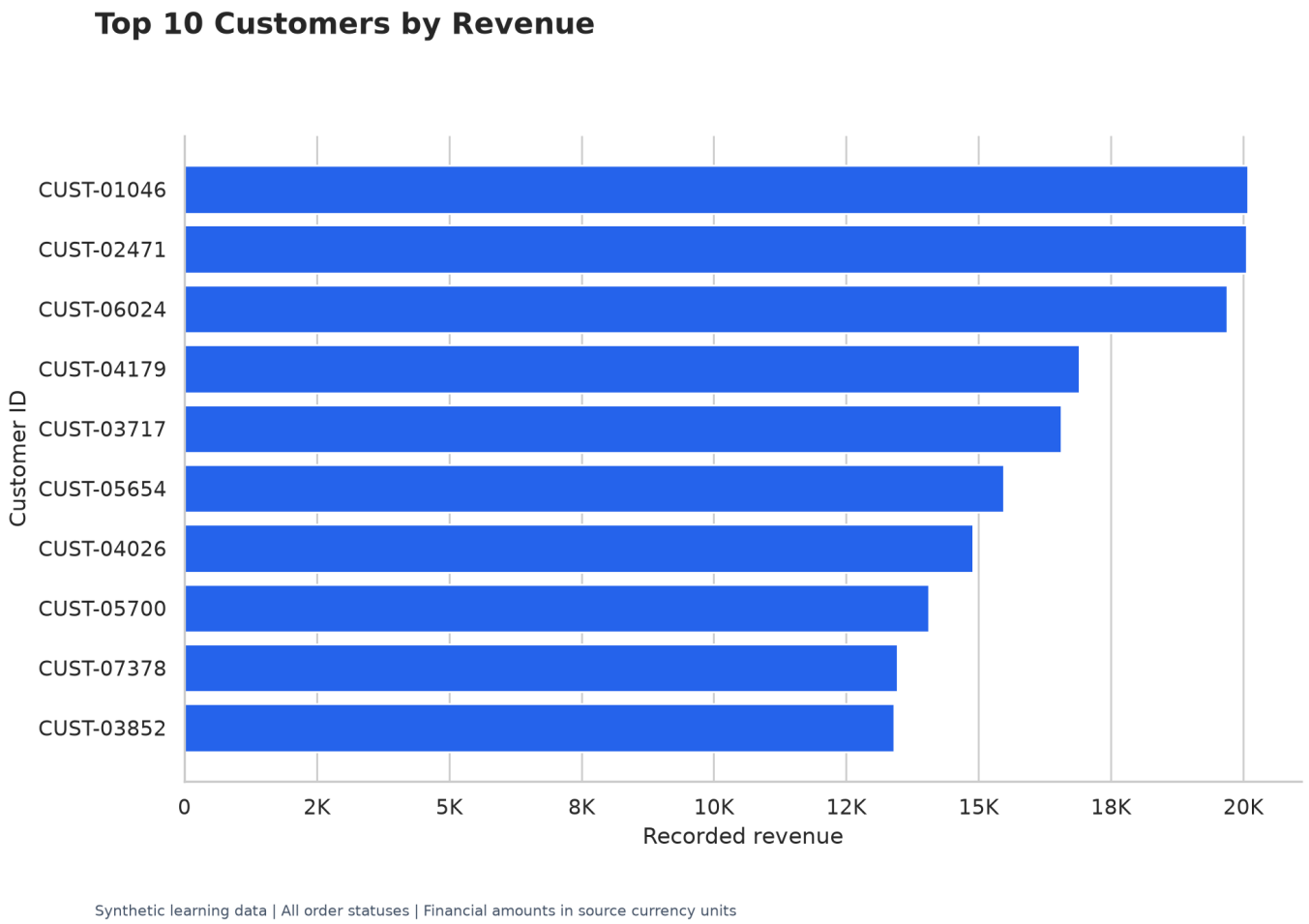
Monthly Revenue Trend



Track recorded revenue by order month. Check monthly order counts and dataset coverage before interpreting increases or declines as changes in business performance.

Analysis scope: All order statuses.

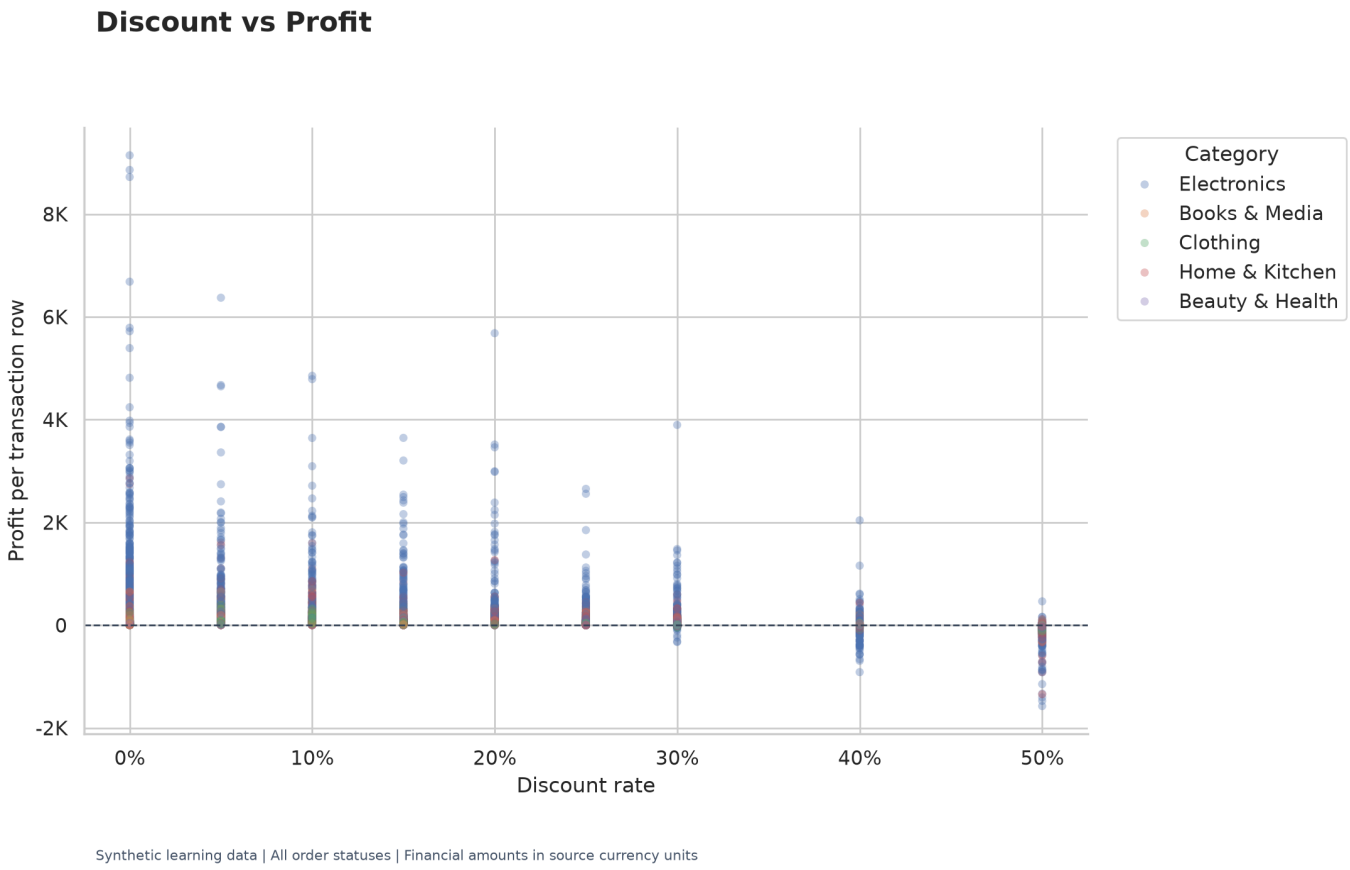
Top 10 Customers



Customers are ranked by recorded revenue and identified by ID. This ranking does not measure customer lifetime value or customer profitability.

Analysis scope: All order statuses.

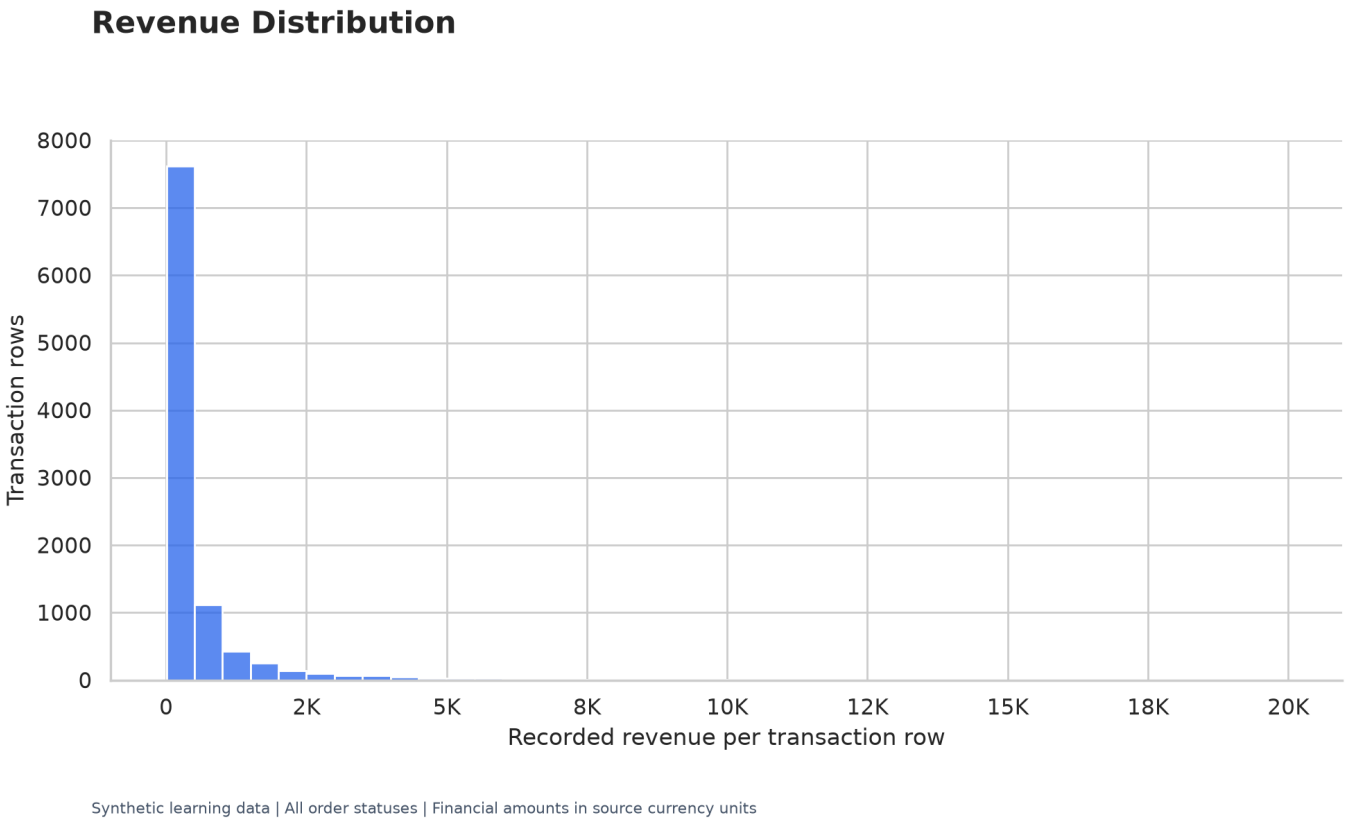
Discount vs Profit



Each point represents a transaction row. Negative values indicate recorded losses. Product mix, transaction size, and cost differences can affect the observed relationship.

Analysis scope: All order statuses.

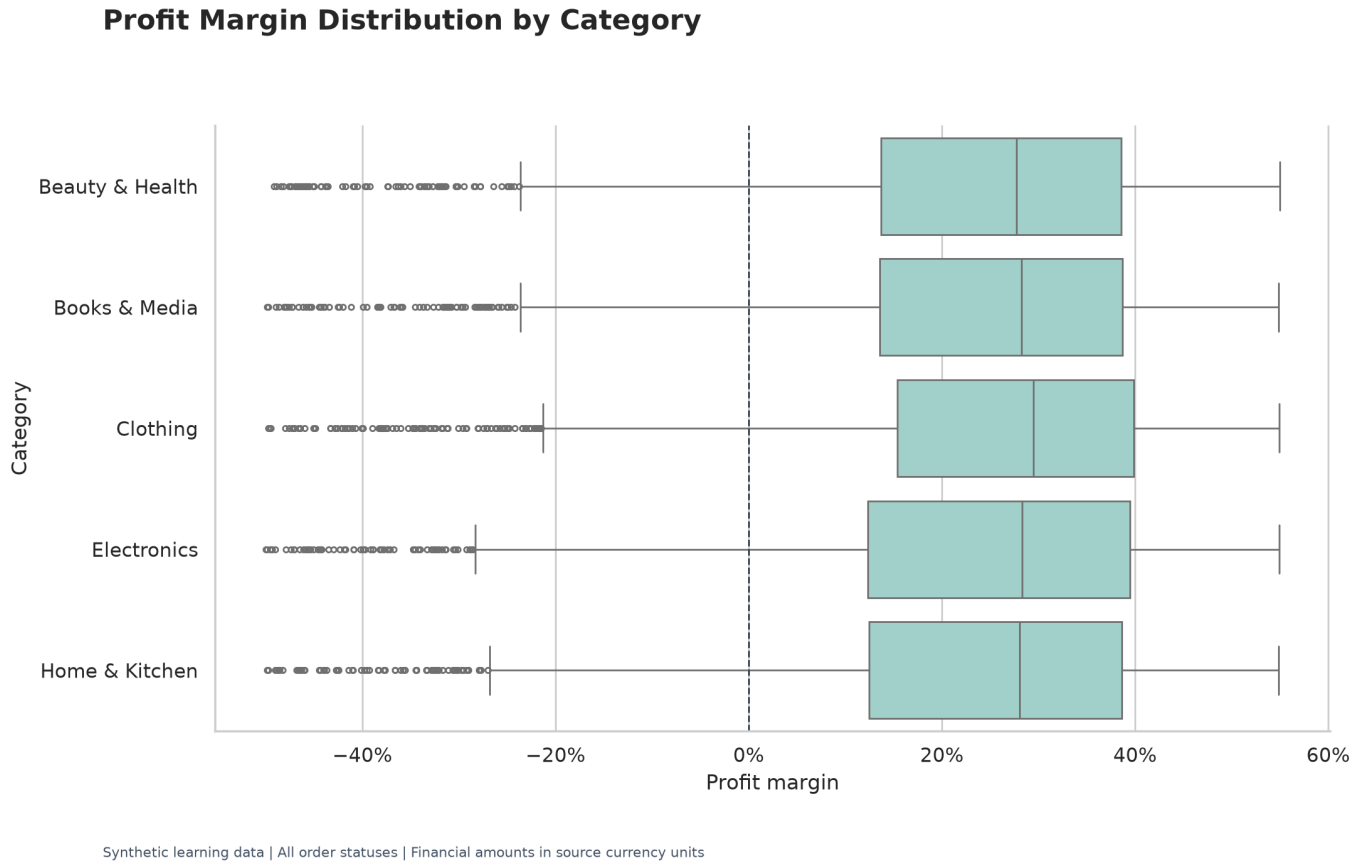
Revenue Distribution



The histogram shows transaction size and skewness. Large transactions are not automatically errors and should not be removed solely because they are unusual.

Analysis scope: All order statuses.

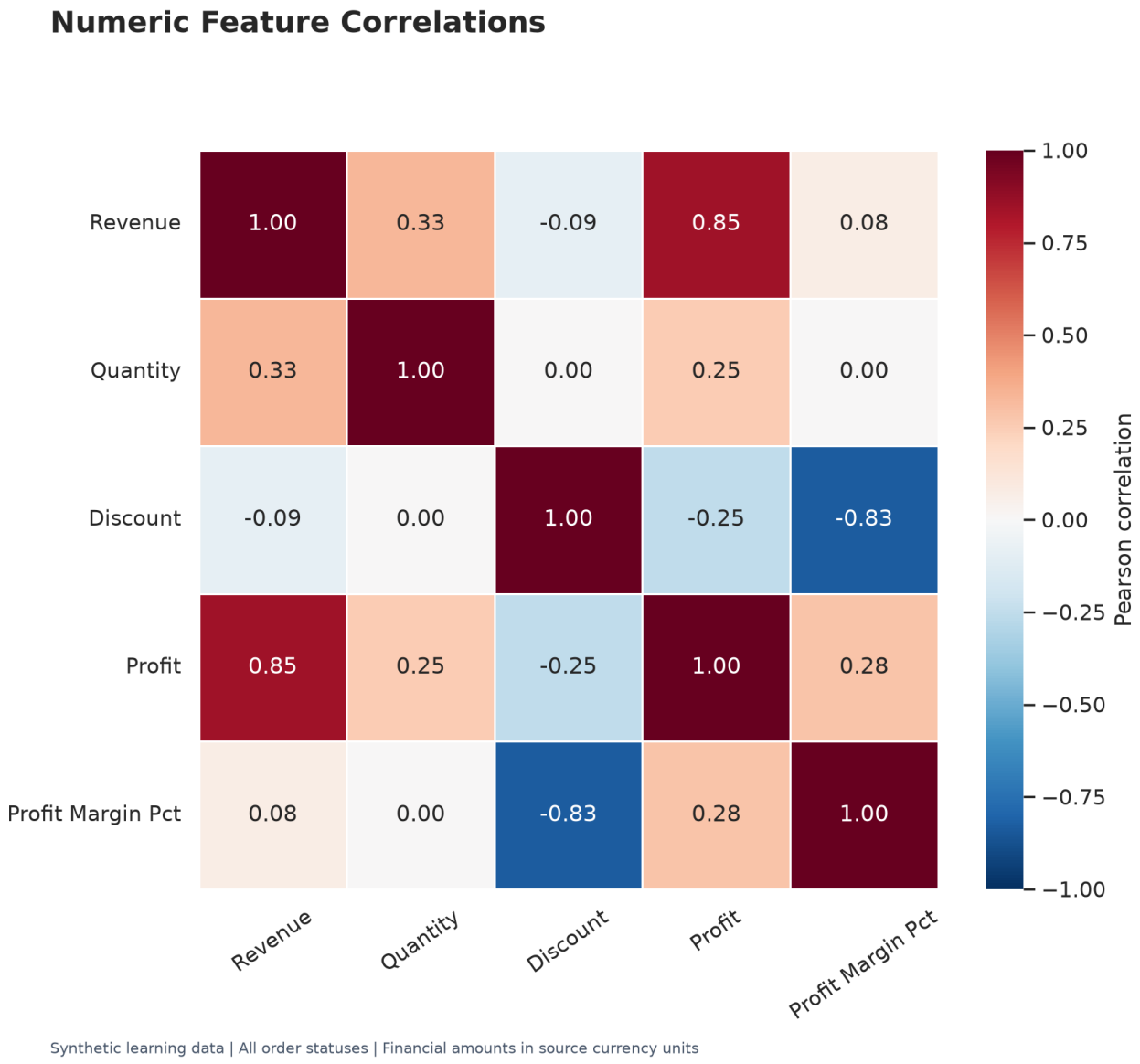
Profit Margin by Category



Boxes show the middle 50% of row margins and the median. Points beyond the whiskers are potential outliers. Row margin distributions differ from aggregate margins.

Analysis scope: All order statuses.

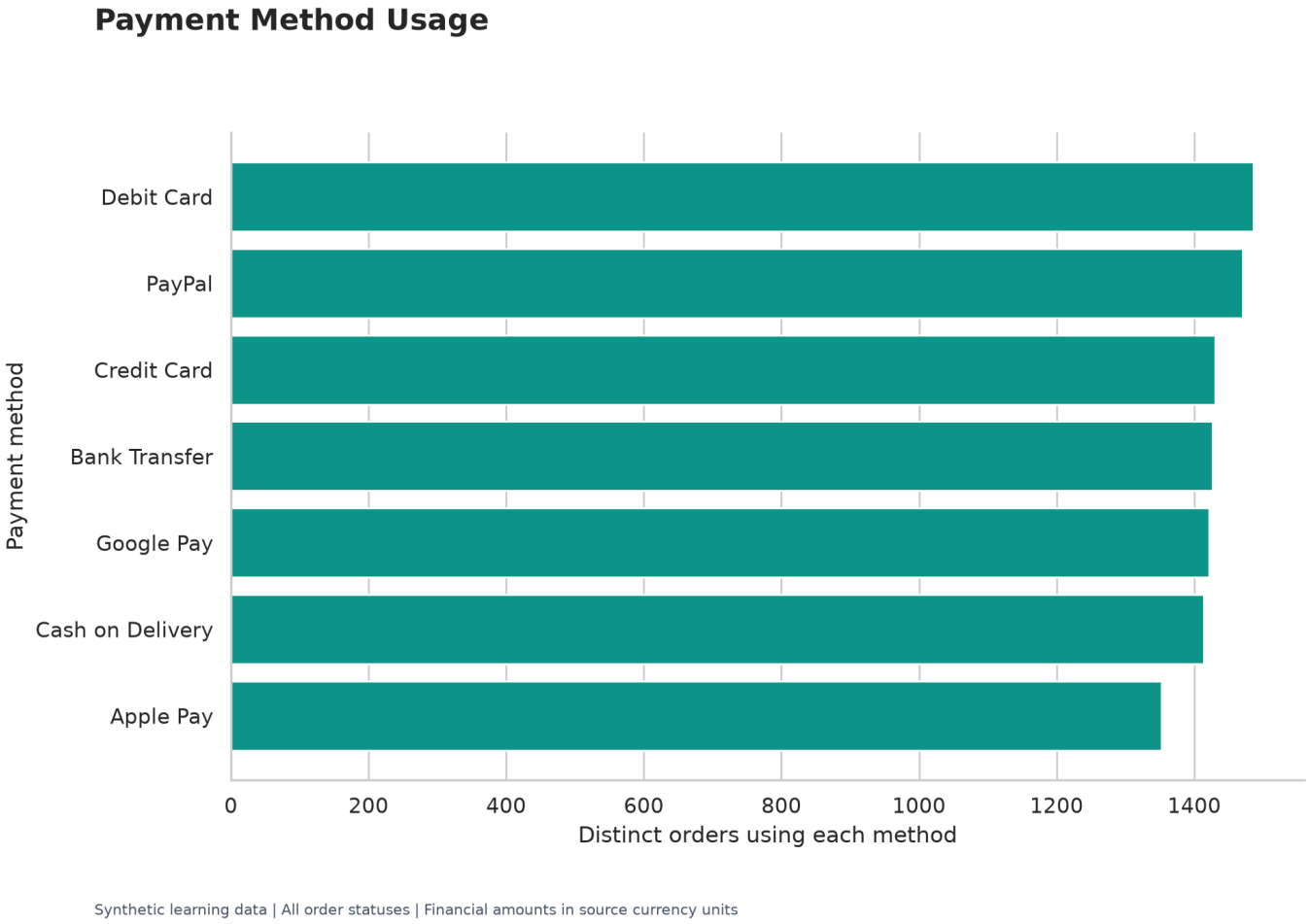
Numeric Feature Correlations



Pearson correlation summarizes linear association. Correlation does not establish causation. Revenue, profit, and profit margin also share mathematical relationships.

Analysis scope: All order statuses.

Payment Method Usage



Usage counts distinct orders using each method. Differences describe this synthetic dataset and should not be treated as evidence of broader consumer preferences.

Analysis scope: All order statuses.

Products with the Lowest Total Profit

Product	Revenue	Profit
Thinking Fast and Slow	10,353.83	2,772.83
The Psychology of Money	11,335.81	2,951.70
Deep Work	11,996.11	3,015.08
Rich Dad Poor Dad	11,462.37	3,049.67
Atomic Habits	12,181.12	3,403.63
Classic T-Shirt	15,792.03	4,403.26
Whey Protein 2kg	18,399.26	4,639.78
Slim Fit Jeans	18,256.18	4,707.26
Chino Pants	18,640.64	4,920.75
Vitamin D3	19,874.70	5,039.37

Lowest total profit does not necessarily mean a product is loss-making. Low sales volume can also produce a low profit contribution. Compare margins, volumes, and costs before deciding on an action.

Recommendations for Investigation

Review costs and discount practices for low-profit products.

Compare discounts within similar products and categories.

Investigate return and cancellation reasons before proposing fixes.

Check monthly order counts and source coverage before explaining the late-period revenue decline.

Assumptions and Limitations

The source Revenue column is treated as recorded sales.

Recorded sales already reflect discounts; do not discount them again.

Discount is a rate: 0.10 represents 10%.

Customers are identified by Customer_ID; names are unavailable.

Average order value uses distinct Order_ID values.

Main revenue and profit totals include all order statuses.

These totals are not delivered-only or refund-adjusted results.

Negative profit values are retained for profitability analysis.

Synthetic data supports demonstration of analytical methods; its patterns are not evidence about an actual business.

Profit is the source's Revenue minus Cost measure. The available data does not establish that it includes every operating expense, tax, or refund.

Unknown customer IDs are excluded from customer rankings. Zero-revenue profit margins are undefined.

Payment counts may overlap if an order uses multiple methods. Status counts may overlap if an order has multiple statuses.

Observed financial correlations partly reflect shared calculation formulas. No causal claims are made.

Conclusion

The project provides a reproducible view of recorded revenue, profitability, customer contribution, and transaction patterns. The findings identify areas for investigation while preserving the dataset's reporting scope and limitations.